

Review article

A survey of smart product-service systems: Key aspects, challenges and future perspectives

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ABSTRACT

The rapid development of information and communication technologies (ICT) has enabled the prevailing digital transformation (i.e. digitalization), where physical products can be readily digitized in the virtual space and seamlessly interconnected. Meanwhile, industries are ever increasingly adopting service business models (i.e. servitization), so as to offer not only physical products but also services as a solution bundle to meet individual customer needs. Such convergence of both digitalization and servitization (i.e. digital servitization) has triggered an emerging IT-driven business paradigm, smart product-service systems (Smart PSS). As a novel paradigm coined in 2014, to the authors' knowledge, only 2 conference papers have provided some literature review to date, and many issues remain uncovered or not comprehensively investigated. Aiming to fill this gap, this paper has conducted a systematic review of Smart PSS or related papers published ever since its first brought up to date (30/06/2019), and selected 97 representative items together with other 37 supplementary works to summarize the tendency towards Smart PSS, its business and technical aspects, current challenges, and future perspectives. From the survey, it is found that several hybrid concerns are the key challenges faced, and self-adaptiveness with sustainability, advanced IT infrastructure, human-centric perspectives, and circular lifecycle management are the core future perspectives to explore. It is hoped that this work can attract more open discussions and provide useful insights to both academics and industries in their exploration and implementation of Smart PSS.

1. Introduction

Nowadays, industrial companies are gradually shifted from a product-oriented business model towards a service-dominant logic [1], i.e. *servitization*, by offering personalized products and services as a solution bundle to meet individual customer needs. This business paradigm is known as product-service systems (PSS) [2,3], where the essence is to deliver other than a product (by ownership), but also its performances (e.g. pay-per-performance) and usage (e.g. pay-per-use) as a bundle [4]. Meanwhile, recent development of information and communication technologies (ICT), such as Internet-of-Things (IoT) and Cyber-Physical Systems (CPS), have enabled a promising market of information densely products, i.e. smart, connected products (SCPs), which have the ability to collect, communicate, process, and produce information [5,6]. Owing to their unique characteristics, SCPs can be readily digitized into the virtual space and remain interconnected with the physical ones, which represents the prevailing digital transformation [7], i.e. *digitalization*. Also, they change the way value created, by embedding IT into

the product itself [5] and acting as the media and tools for generating value-added e-services as a solution bundle, which is known as *digital servitization* [8,9]. From system perspective, such emerging IT-driven business paradigm is named Smart PSS, which was first coined in a conference paper published in 2014 (Valencia et al. [34]), and **widely acknowledged in the journal publication in 2015 as "the integration of smart products and e-services into single solutions delivered to the market to satisfy the needs of individual consumers."** [11].

Ever since its introduction, similar terms such as Digitalized PSS [8], Cyber-Physical PSS [12] have also been proposed, and many related research works have been introduced in this research area as well. However, to the best of authors' knowledge, only 2 conference papers [13] and [14] have reviewed the Smart PSS from either business aspect or technical aspect, respectively. No comprehensive survey has been given in this field, let alone to point out the transformation of such IT-driven PSS, and its current challenges, and future perspectives. Aiming to fill the gap, this paper has undertaken a systematic survey of the recent publications since 2014, selected 97 highly related journal/

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conference papers to summarize the key aspects of Smart PSS, and further outlined the main challenges and prospective research directions in this field. The rest of this paper is organized as follows: [Section 2](#) describes the evolution of PSS in line with the IT innovations. [Section 3](#) gives a systematic literature review process to select the relevant publications and conduct a statistical analysis accordingly. Based on the selection, the key aspects of Smart PSS are elaborated in [Section 4](#), considering both business and technical perspectives. Moreover, the main challenges and future perspectives are highlighted in [Section 5](#) and [Section 6](#), respectively. The major contributions and limitations of this survey is given at last.

2. From PSS to Smart PSS

Following the definitions in our previous work [15], the IT-driven PSS evolution can be divided into three phases ever since PSS first coined in 1999 [16], beginning from *Internet-based PSS* (1999 -) (conventional PSS) [17], *IoT-enabled PSS* (2010 -) [18], to today's *Smart PSS* (Valencia et al. [34]), as shown in [Fig. 1](#). The vertical axis represents the levels of digital servitization of each phase, while the horizontal axis stands for the timeline of the IT innovation. The key technologies and significant academic milestones of each phase are listed in [Fig. 1](#) and elaborated below.

Internet-based PSS benefits from the wide usage of Internet/Mobile Internet. It follows the basic principles of PSS as a product and e-service bundle in the Internet environment, and the major concern still lies in the efficient delivery of data/information (e.g. 2G/3G/4G) with little digitalization consideration. In this phase, the first publication can be traced back to 1999, when web-based computer-aided maintenance was proposed by Lee and Wang [19]. In 2003, they further introduced the first concept of “smart product and service systems” [17] by leveraging the Internet to provide e-maintenance services. Although that Smart PSS largely differs from today's one, it is interesting to mark their novelty and contribution to future works. Meanwhile, other key milestones such as service-dominant logic (S-D Logic) [1], and eight types of PSS business models [20] also set the foundation of servitization.

IoT-enabled PSS is triggered by the prevailing adoption of IoT [21], as the key technology to drive the digital transformation [22]. IoT-enabled PSS [18] was first introduced by exploiting RFID devices in the industrial environment, where RFID readers/tags (product) and their embedded information are connected to the Internet for production maintenance purposes (digital service). Moreover, platform approach was adopted as an organizational business perspective to leverage the value of IT-enabled interactions and digital technologies based on modularity [23,24]. Hence, in this phase, the major concern exceeds the efficient delivery of data/information as in the Internet-based PSS, but to take both online information processing capability and modular service composition (cloud services) into consideration for digitization and servitization purposes.

Smart PSS are enabled by the wide adoption of SCPs [25], of which IT is embedded in the product itself for value creation. Meanwhile, it is also motivated by the digitalization of products and services enabled by the CPS [12] and Big Data technologies [26], where the digital twin [27] established between components allow autonomous interaction and further optimization of them, which represents the digital servitization [8]. Hence, in this phase, despite the ubiquitous connectivity and online intelligence enabled by the IoT-enabled PSS, both cyber space (online) and physical space (offline) should also be considered to realize the Smart PSS. Furthermore, the degree of “smartness” can be further divided into five levels (i.e. smart, connectivity (I), smart analytics (II), digital twin (III), cognition (IV) and autonomous (V)) according to the 5C architecture [28,29].

However, to the authors' knowledge, some articles argue that Smart PSS starts from 2010 onwards [30,31]. It is partially agreed since the adoption of IoT marks the infancy stage of digital transformation [7], and serves as the prerequisite for Smart PSS (i.e. ubiquitous

connectivity). However, this work respects the truth when the terminology was first coined in 2014 and widely accepted thereafter, with many new technologies exploited other than IoT. Hence, to avoid any bias, the authors treat the IoT-enabled PSS as an extended scope of Smart PSS, as shown in [Fig. 1](#).

3. Systematic literature review process

3.1. Selection of literature

The systematic literature selection process is depicted in [Fig. 2](#). The authors have carried out the first-round basic search using the Scopus and Web of Science (WoS) databases, owing to its wide coverage of all the major peer-reviewed articles in academia.

The search sentence is written as: “Topic = (PSS OR “PRODUCT-SERVICE SYSTEMS” OR SERVITIZATION) AND Topic = (SMART OR DIGITAL); Time Span: 2014 – 2019; Language: English”. The time span matches well with the first introduction of Smart PSS and afterwards. After the initial screening, 384 items from Scopus and 589 from WoS were found (accessed on 30/06/2019). Then, a second-round search was conducted, where only academic journals and conferences were included while the topics not within the field are excluded (e.g. optics, medical) to refine the existing papers, and 198 relevant items are extracted. Then, a more detailed review was undertaken by reading the content and references of the items, and those ones only discuss smart services are excluded (e.g. e-government, e-retailer), while 6 related ones (e.g. digital platform, value co-creation) found in the references are added (including 1 significant editorial material [32]). Finally, after more in-depth reading of those ones, a total of 97 items are selected as the foundation of this survey.

3.2. Statistical analysis of selected items

A general statistical analysis is conducted based on the 97 selected items, including the number of publications in each year, country of origin of the leading institution, and publication name of journals or conferences, as shown in [Fig. 3](#). It can be seen from [Fig. 3\(a\)](#) that there is an increasing concern/discussion of Smart PSS during the past 5 years, especially in the recent 2 years. This can also be revealed by the Special Issue on Stakeholder-oriented Design Innovation of Smart Product-Service Systems launched in the journal of Advanced Engineering Informatics* in 2019 and the most recent 11th CIRP Conference on Industrial Product -Service Systems, of which the conference theme is entitled “Smart Product and Service Twin”†. Meanwhile, among all the selected publications, most are from European (e.g. German (16), UK (9), Sweden (8), Italy (8)), Asian countries (e.g. China (12) and Singapore (7)) and USA (7), as shown in [Fig. 3\(b\)](#). The result is in line with the fact that Smart PSS was originated from Europe, and may somehow reflect the digital servitization status in those respective countries. Moreover, from [Fig. 3\(c\)](#), one can find that CIRP is the major publisher in this area, including 17 conference papers published in Procedia CIRP (15 from CIRP Conference on Industrial Product -Service Systems, and 2 from CIRP Conference on Manufacturing Systems) and 7 journal papers in the CIRP Annals – Manufacturing Technology. Other published journal/conference papers are dispersed in various research fields, including environmental engineering (e.g. Sustainability, Journal of Cleaner Production), industrial engineering (e.g. IEEE Transactions on Industrial Informatics, Journal of Intelligent Manufacturing), business and economics (e.g. Strategic Change, Industrial Marketing Management, Harvard Business Review), and artificial intelligence (AI) (e.g. Advanced Engineering Informatics), which reveal its interdisciplinary essence.

* <https://www.journals.elsevier.com/advanced-engineering-informatics/call-for-papers/special-issue-on-stakeholder-oriented-design-innovation>.

† <http://www.cirpips2.org/>.

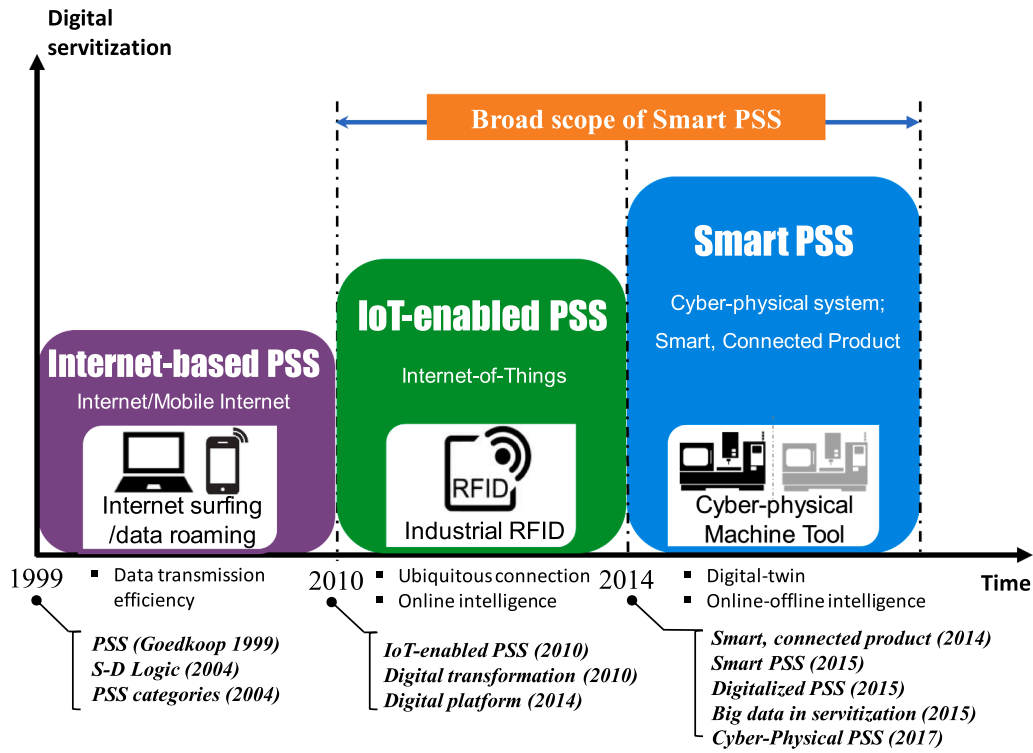


Fig. 1. An overview of IT-driven PSS evolution.

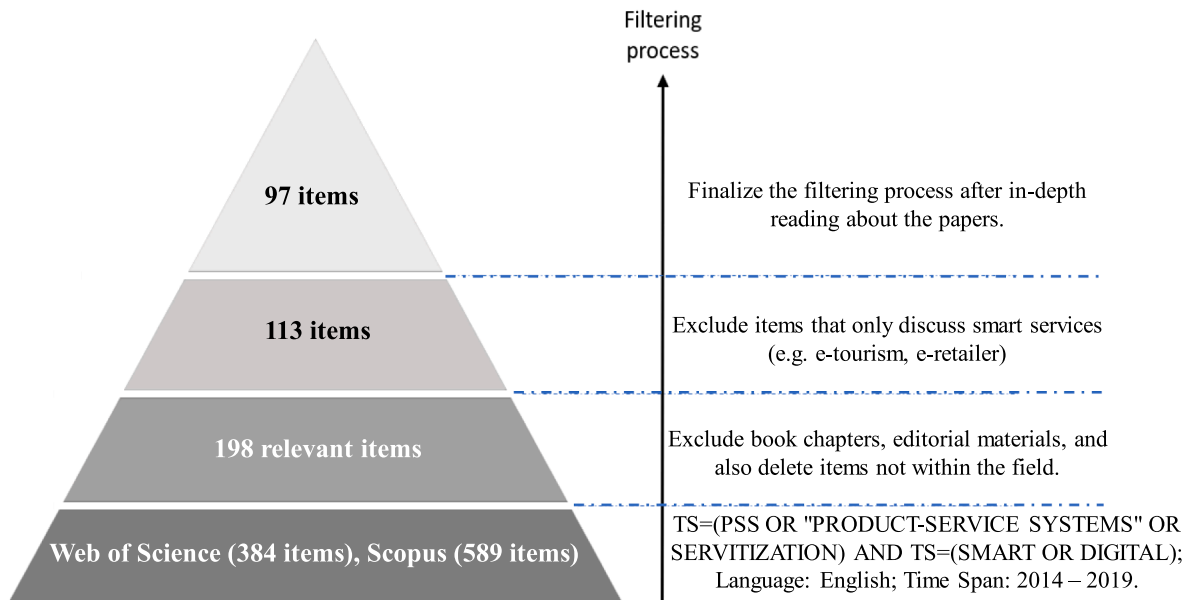


Fig. 2. The systematic literature selection process.

3.3. Organization of the survey

To better organize the survey, key aspects of the selected items can be further classified into three parts, i.e. *theories*, *methodologies*, and *applications*, as shown in Fig. 4. *Theories* part introduces the fundamentals of Smart PSS, including the definitions and key elements (Section 4.1), and its enabling technologies (Section 4.2). *Research methods* elaborates both the technical aspects adopted in the Smart PSS development process (Section 4.3) and implementation process (Section 4.4), engineering lifecycle adoption (Section 4.5) and business innovation aspects (Section 4.6). *Application scenarios*, depicts the real-life applications of Smart PSS (Section 4.7). Based on that, existing

challenges (Section 5) and future research perspectives (Section 6) can be derived as *research outcomes* of this comprehensive survey.

4. Key aspects of Smart PSS

Based on the systematic literature review process, this section outlines the key aspects of Smart PSS, including its definitions and key elements, technical aspects (i.e. enabling technologies, development and implementation approach, engineering lifecycle management), and business innovation aspects (i.e. business model, digital platform, value co-creation, digitalization capability and circular economy).

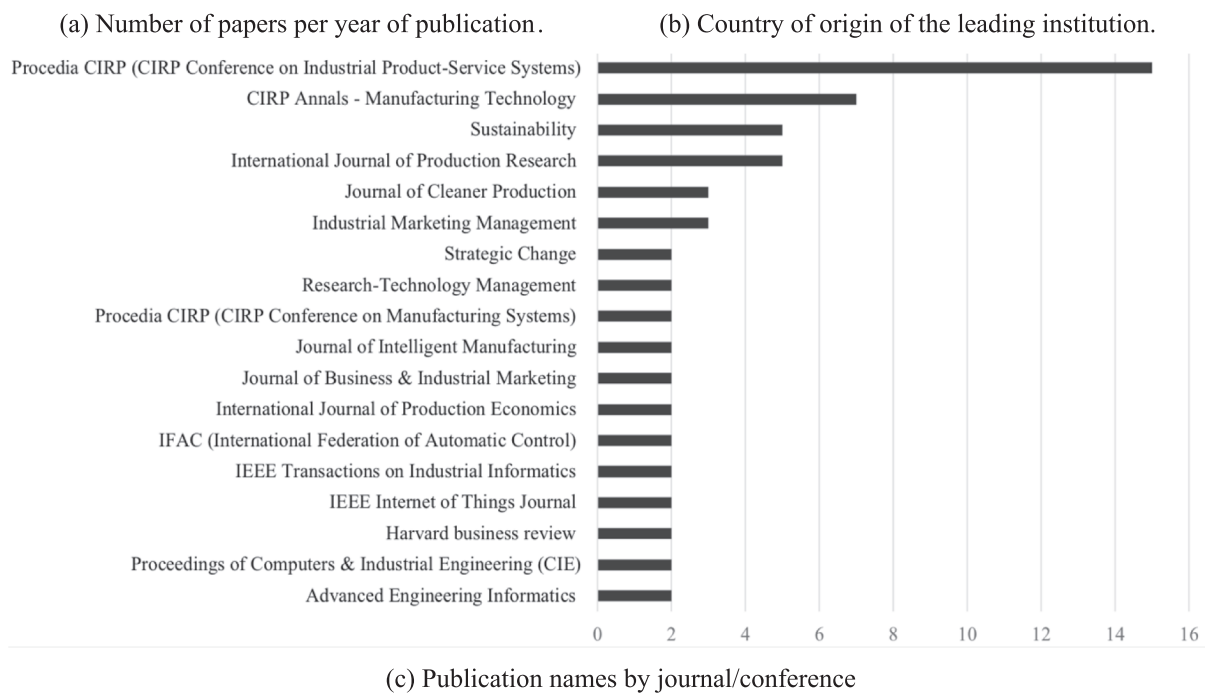
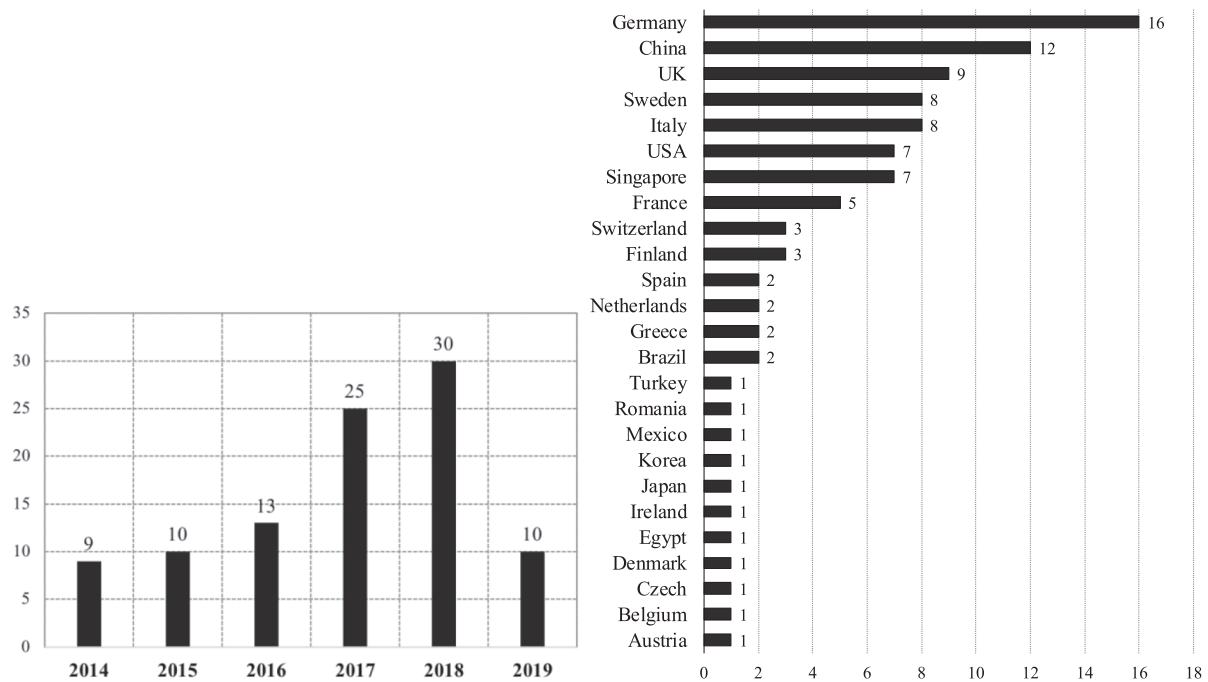


Fig. 3. The statistical analysis of the selected publications.

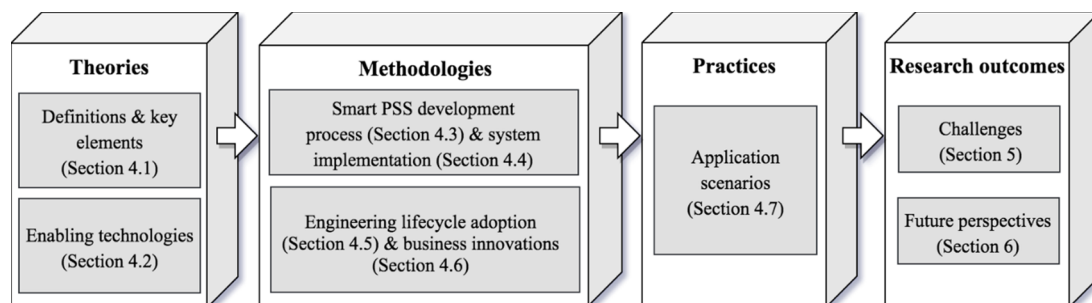


Fig. 4. The overall structure of the survey.

Table 1
Smart PSS definitions and key elements.

Smart PSS related terms	Reference	Definitions	Key elements
Smart PSS	Valencia et al. [34,35]	"Smart products and its generated e-services into a single solution by embracing disruptive ICT."	Organization-wise: stakeholder involvement; Design-wise: complex market offerings, ICT technologies, e-services, interactions between the Smart PSS and end-users; context, and a life-long development issue
	Kühlenkötter et al. [36]	"A digital-based ecosystem of value creation characterized by high complexity, dynamics and interconnectedness among stakeholders."	/
	Zheng et al. [15,37,38]	"An IT-driven value co-creation business strategy consisting of various stakeholders as the players, intelligent systems as the infrastructure, smart, connected products as the media and tools, and their generated services as the key values delivered that continuously strives to meet individual customer needs in a sustainable manner."	SCPs and generated e-services Stakeholders
	Liu, et al. [39]	"A platform service ecosystem, in which platform is made up of smart products and smart services, while multiple service systems constitute a service ecosystem."	Intelligent system (smart environment)
Smart industrial product-service system of systems	[40]	"A consolidation of resources on people, intelligent machines and devices, infrastructures and networks to fulfil customer needs as problems to be solved."	Service platform, SCPs and smart services
Smart Product Service Ecosystem	[13]	"An ICT based dynamic ecological Smart PSS network, which integrates customers, smart product service systems, smart service platform and product service suppliers for value co-creation and customer experience improvement, by means of smart interaction, mutual cooperation, resource sharing and optimal configuration."	From the perspective of ecosystem: (1) connection, (2) business model, (3) relationship, (4) user experience, (5) technologies, and (6) marketing

4.1. Definitions and key elements

Among existing literature, the major definitions and key elements of Smart PSS are listed in Table 1, including both system level and eco-system level (system-of-systems). One can find that, although lacking a unified definition, it is widely accepted that SCP and its generated digital services are the fundamental composition of the Smart PSS, and it undertakes an IT-driven value co-creation manner to fulfil customer needs. Besides those definitions, similar terminologies are also summarized, as shown in Table 2, including cyber-physical PSS [12], PSS-CAD [33] and digitalized PSS. All these three terminologies share the same fundamentals as Smart PSS, i.e. digitalization and servitization enabled by the advanced information and digital technologies. Nevertheless, each one only emphasizes a specific scope or level without comprehensive review.

Based on the literature, Fig. 5 classified extracted key elements of Smart PSS into three levels, i.e. *product-service level*, *system level* and *system-of-systems level*. *Product-service level* refers to a single solution bundle consisting of SCP, its generated services and supporting infrastructures to meet individual user needs. *System level* expands the scope from the functionality of a discrete solution to the systematic integration of a family of solutions, the involvement of various stakeholders (e.g. users, service providers) and platform-based intelligent systems to optimize the overall performance. *System-of-systems level* further extends the industry boundaries to a set of associated Smart PSSs as well as the related external auxiliary systems (e.g. data management system) to make the greatest impact on the total ecosystems, such as a smart home, a smart farm or a smart building [5].

4.2. Enabling digital technologies (digitalization capabilities)

From technical aspect, to realize Smart PSS, the enabling technologies are counted from the 97 selected items based on their occurrence (see Fig. 6(a)) and the digitalization capabilities can be further classified into three categories, i.e. *connect capability*, *intelligence capability*, and *analytic capability* [41–44], and each prior category serve as the foundation for the latter one, as shown in Fig. 6 (b).

Connect capability stands for the ability to connect various SCPs through wireless communication networks. IoT (37) serves as the fundamental technology to enable ubiquitous connectivity (connectedness), so as to enable manufacturers or service providers to collect data and offer product related services based on the product operation data [6,45]. Wireless communications can be achieved through radio-frequency identification (RFID), Bluetooth, Global positioning system (GPS), wireless sensor network (WSN), product internet (PI) and etc.

Intelligence capability denotes the ability to configure hardware components to sense and capture information with less human intervention. It is enabled either by the technologies to upgrading hardware components with smart subcomponents, such as embedded system, smart sensors and actuators, or the technologies to collect and process massive information of the product-sensed data or user-generated data, such as cloud computing (18) [42] and edge computing (1) [46].

Analytic capability represents the ability to transform the data/information available at hand into valuable insights and actionable directives for the company which is enabled by the Augmented Reality (4) (e.g. remote operation access [47]), CPS (25) (e.g. digital twin [48]), AI (19) (e.g. machine learning for predictive maintenance [49]) and Big Data analytics (12) (e.g. Big Data in servitization [50]).

4.3. Smart PSS development process

A systematic Smart PSS development process is of paramount importance to the final success of smart product-service provision, of which the main approaches are outlined in Table 3. One can find that the main focus lies in two aspects: (1) data-driven platform-based approach; (2) value co-creation process, as depicted in Fig. 7. For the prior

Table 2
Other similar PSS terminologies.

Other PSS terms	Reference	Definitions
Cyber-physical PSS	[12]	"Manufacturers at this stage not only provide complex PSS to their customers, but also incorporate ICT solutions as a novel component in the product-service bundle, creating intelligent, independent operating systems that deliver the highest level of availability possible and optimize operations while reducing resource inputs."
Digital PSS (PSS-CAD)	[33]	"Computer-aided design (CAD) system to support the design of PSS."
Digitalized PSS	[8]	"A combination of CPS technology with PSS business models."

one, it is discussed from the product-service provision generation perspective [51,52], where massive user-generated and product-sensed data are collected mainly through SCPs and further analyzed in the service platform for Smart PSS provision generation. For examples, Zheng et al. [15] addressed a data-driven digital twin-enabled servitization process, and Scholze et al. [53] extended the scope with the

consideration of knowledge management, context modelling and security configuration, improving the quality and reliability of the generated product-service provision. For the latter one, it is investigated from a value-driven perspective. For instances, Liu et al. [39] proposed a four-stage value co-creation process, of which both service providers and users are actively engaged in the so-called interactive activity

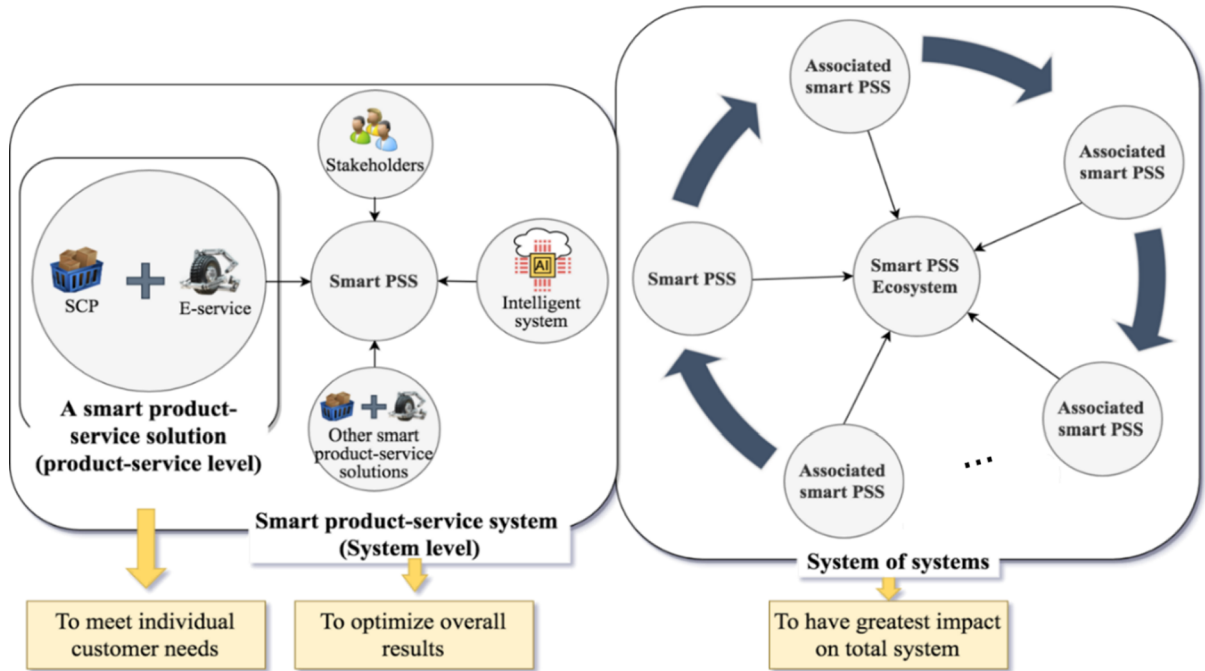


Fig. 5. An overview of Smart PSS from product-service level, system level, and system-of-systems level.

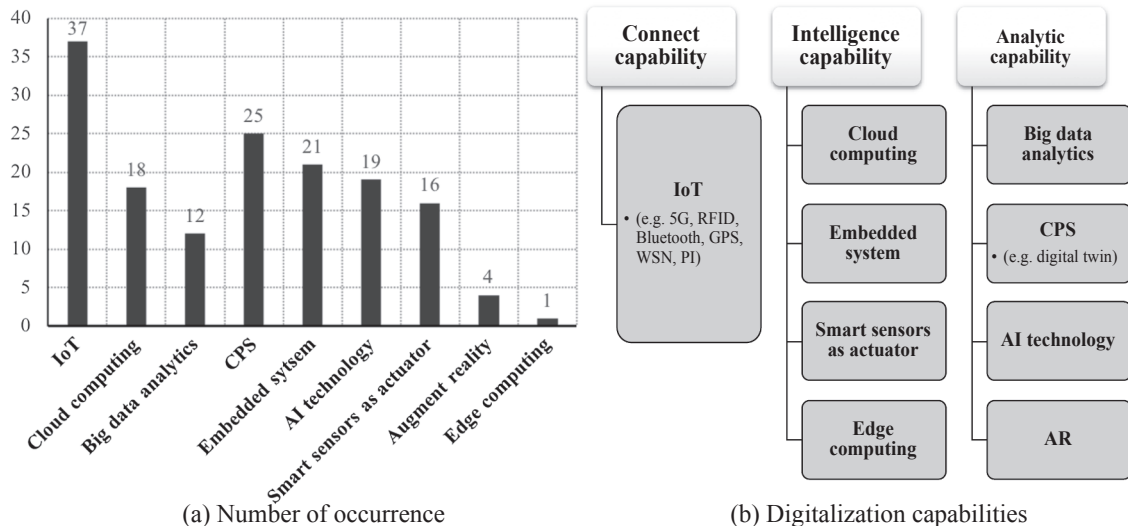


Fig. 6. Enabling technologies of Smart PSS derived from the selected papers.

Table 3
Examples of Smart PSS development process.

Task	Ref.	Smart PSS development process
Smart PSS development process	[15]	(1) Platform development, (2) data acquisition and pre-processing, (3) data analytics for service innovation, and (4) digital-twin-enabled service innovation
Novel PSS development process	[53]	(1) Idea creation, (2) social simulator, (3) knowledge specification/data mining, (4) data capture/cyber physical features selection, (5) functional specification determination, (6) context modelling, (7) security configuration, (8) product extension services orchestration, and (9) PSS engineering system development.
Smart service development	[54]	(1) Data collection from smart equipment, (2) customer needs assessment, (3) data analytics and decision support, (4) business model and value creation, and (5) smart service selection
Value-oriented Smart PSS development	[39]	(1) Co-exist the value flow of stakeholder, (2) co-design the value proposition, (3) co-implement the interactive value, and (4) co-evaluate the performance value.
SPSE development process	[13]	(1) Define (demand system boundary and customer demands), (2) discover (discover typology model, robust mechanism, value emergency), (3) design (design product clustering, service flow and service recommendation), and (4) delivery (capability planning, process management and performance assessment)
Smart PSS design process	[55]	(1) Problem definition (analyzing the service context and product characteristics), (2) resolution generation (generating service resolutions the smart PSS solution), and (3) resolution design (smart PSS modeling).

diagrams. Meanwhile, Parida, Sjödin and Reim [32] articulated the components of value creation, value delivery, and value capture in the digital servitization process.

4.4. Smart PSS implementation architecture

As an IT-driven paradigm, the implementation architecture of Smart PSS sets the foundation [56]. Table 4 summarizes the typical system architectures of Smart PSS from the selected items. Nevertheless, only three articles analysed its implementation architecture, two of which are in the context of Smart PSS [39,46] and the other is an IoT-driven service framework [57]. Nevertheless, the organization structures and strategic innovation [58,59] are still lack of sufficient study in the existing system architectures. Based on the development process depicted in Section 4.3 and the typical system architectures listed in Table 4, a generic system architecture, including *physical resource layer* (large-scale SCPs as data collectors), *networked platform layer* (IoT or PI environment for ubiquitous connectivity), *data management layer* (cloud-edge computing for information processing), *application layer* (service composition for knowledge management), and *business intelligence layer* (intelligent decision making to delivery values) for Smart PSS implementation can be presented in Fig. 8, which follows the DIKW (data-information-knowledge-wisdom) model [41]. Moreover, tackling safety and security challenges of Smart PSS should also be emphasized [60], which is beyond discussion here.

4.5. Adoption in engineering lifecycle management aspect

From technical aspect, this section summarizes the existing adoptions of Smart PSS in different engineering lifecycle stages, including *design stage*, *manufacturing stage*, *distribution/logistic stage*, *usage stage* and *end-of-life stage* [28,61], which are listed in Table 5 and elaborated below.

4.5.1. Design stage

Tasks in design stage can basically classified into two categories, namely *requirement management* and *smart design*. The literature about the *requirement engineering* in Smart PSS mainly focus on the requirement capture and evaluation, in which customer demands and feedbacks are collected in a data-driven manner not only in the early design stage, but also in manufacturing stage [36] and usage stage [62,64]. Designers can identify and quantify the customer demands based on large-volume user behaviour and demands data instead of relying solely on their experiences [64]. For instances, Gupta et al. [65] developed a systematic approach for gathering requirements during production through the remote access by AR to examine the partially or fully assembled aircraft and its modules. Wang et al. [62] proposed a data-driven and graph-based methods for implicit requirement extraction during usage stage in Smart PSS.

Smart design contains product and service development/innovation. From product-oriented aspect, four levels of the SCP capabilities are addressed by Porter and Heppelmann [5], namely monitoring, control, optimization and autonomy. To realize the autonomy capability, some features should be attached to the smart products (SCPs), e.g. self-awareness [68] and reconfigurable [88]. Filho et al. [68] reviewed the state-of-the-art of self-aware smart products and defined three degree of achieving self-aware smart products, including information handling, intelligent item and intelligence through network. Savarino et al. [88] further underlines the context-specific reconfiguration of smart products during their usage phase. Meanwhile, from service-oriented perspective, it includes both e-services (product-independent) and digitalized services (product-dependent) generated from the SCPs. For the former one, Verdugo Cedeño et al. [54] developed a customer-oriented approach for developing smart e-services based on IoT and other techniques. Lim et al. [70] proposed a systematic data-driven approach to develop vehicle operations and health management e-services. For the latter one, advanced digitalized services (e.g. digital twin [48,67]

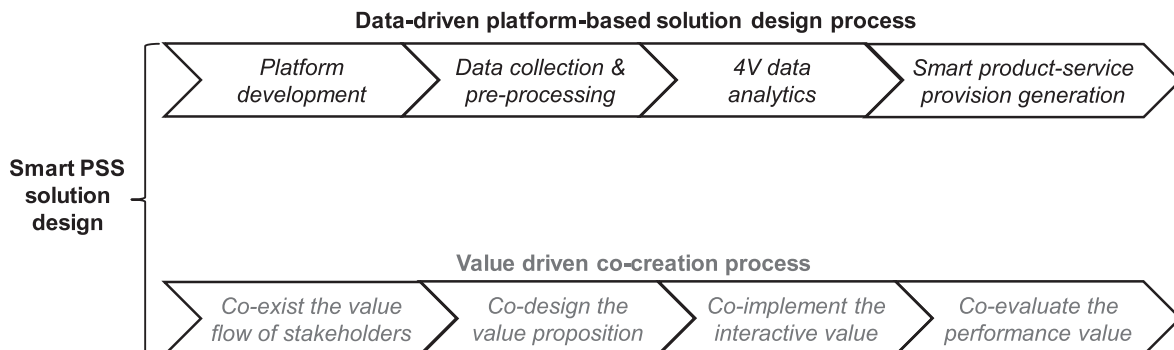


Fig. 7. Smart PSS development process.

Table 4
A summary of system architectures.

System	Reference	System architecture of Smart PSS or similar systems
Smart PSS	[39]	L1: Smart devices (embedded OS and physical components); L2: Network (transports of network equipment and transmission standards); L3: Data management (data storage and processing); L4: Applications (connect and interoperate the service platform with application programs)
Smart PSS	[46]	L1: SCP layer (a number of smart devices connected in an IoT environment); L2: Edge computing layer (local data streams processing, and service matching/providing from the cloud to users with low latency); L3: Software defined network management layer. L4: Cloud computing layer (big data analytics with high latency)
An industrial IoT system architecture-based services framework	[57]	L1: Edge tier (raw data collected from IoT devices, sensor data storage locally and edge implementation of services can be applied to this data); L2: Platform tier (services composition for operating in the cloud); L3: Enterprise tier (user access to cloud services' configuration, data, and outcomes)

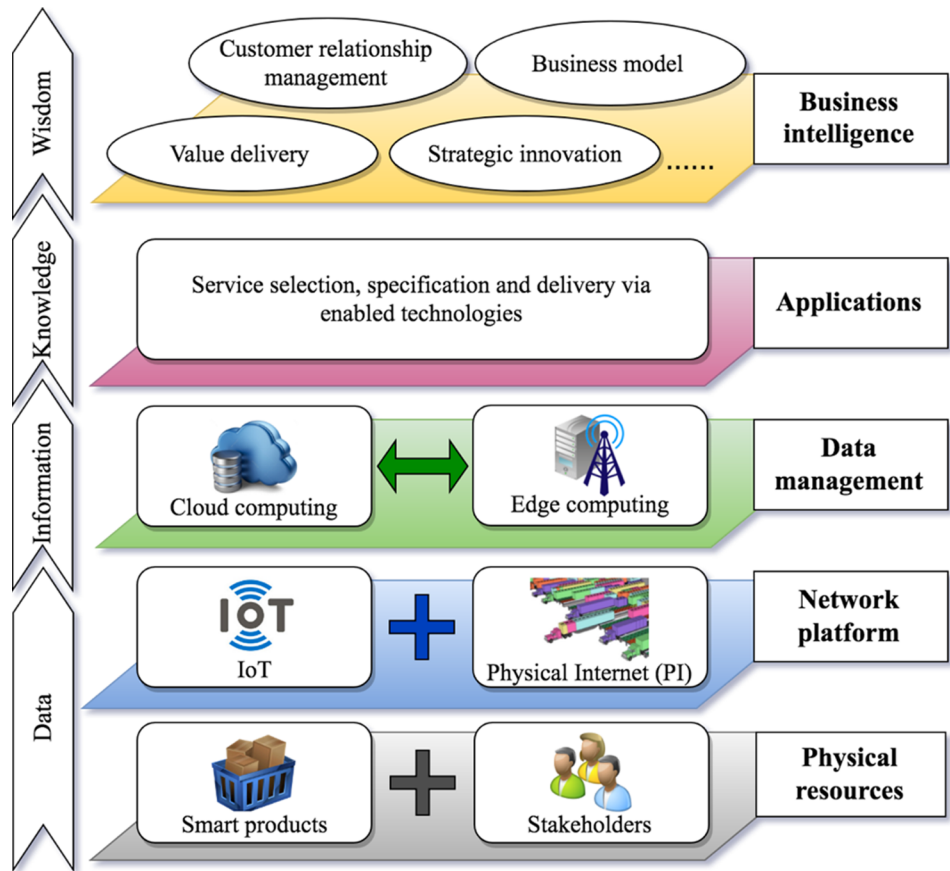


Fig. 8. A derived system architecture for Smart PSS implementation.

and AR [65]) have the ability to offering more accurate decisions for designers in the conceptual design, detailed design and virtual specification in the design stage.

4.5.2. Manufacturing stage

CPS (e.g. digital twin) [89,90], as unique types of Smart PSS, are the enabling technologies widely adopted in the manufacturing process, especially in the *smart production* and *smart inspection*. For the prior one, it mainly contains resource management, production plan and production process control/reconfiguration [48]. Tao et al. [48] discussed the resource management plan with the support of digital twin-based shop floor, which serves as the basic performer to organize resources for the specific product. Uhlemann et al. [75] addressed that the digital twin can support the simulation-based production process optimization. Furthermore, Tao et al. [48] also discussed digital twin-based production process monitoring and control, where the shop flow adjustment can be generated by the Shop Floor Service system (SSS) based on the physical disturbance data, simulation conditions data, environment

data, etc. For the latter one, it aims to minimize the geometrical variations in the final product. Both Söderberg et al. [77] and Schleich et al. [67] suggested that digital twin can realize real-time control and optimization of product quality during design and production stage. One may also refer to [91,92] for more details.

4.5.3. Distribution/logistic stage

Global scope of logistic, more transparent real-time information trajectory and more sustainable services can be achieved with the adoption of Physical Internet (PI), PSS and cloud computing [78]. A seamless information flow between customers, partners and organizations can be ensured by developing the concept of cloud-chain proposed by Vazquez-Martinez et al. [79]. Meanwhile, the privacy of the product delivery services can also be guaranteed by using encrypted packs. Hence, the quality of logistic services is significantly increased in the context of Smart PSS.

Table 5
Summary of adoptions in engineering lifecycle management.

Stage	Task category	Specification	References
Design stage	Requirement management	Requirement capture & evaluation	[13,40,62–66]
	Smart design	Product development	[15,28,33,48,64,65,67–69]
Manufacturing stage	Smart production	Service development/innovation	[15,54,69–73]
		Resource management	[48,74]
	Smart inspection	Production planning	[48,64,67,68,75]
		Production process control/reconfiguration	[48,64,76]
Distribution /logistic stage Usage stage	Smart logistic	Geometry assurance	[67,77]
	Smart operation/maintenance	Logistics and packaging	[64,78,79]
		Performance assessment	[28,29,48,70,80]
	Smart reconfiguration	Monitor, maintenance, diagnosis	[29,30,47–49,57,64,81–84]
		Engineering change management	[38,85]
End-of-life stage	Smart reuse	Product reconfiguration	[86,87]
		Product reuse	[30]
	Smart remanufacturing	Product remanufacturing	[30]
	Smart recycling	Product recycling	[30]

4.5.4. Usage stage

Usage stage is dramatically changed by the Smart PSS adoptions, and most widely discussed among the selected items, of which can be further divided as *smart operation/maintenance*, *smart reconfiguration*, *smart reuse* and *smart remanufacturing*.

Smart operation/maintenance mainly contains two types of specific tasks, i.e. performance assessment and monitor/maintenance/diagnosis, as listed in Table 5. Performance assessment mainly focuses on the analytics of the PSS offerings, such as the running time, costs and operational flexibility, while the tasks of monitor/maintenance/diagnosis highlight the monitoring and actuations to the PSS offerings [30]. Lots of advanced enabling technologies including AR [47,81], digital twin [82], modular sensing platform [83] and ontologies [57] are applied in this kind of tasks, giving new capabilities to the PSS.

Smart reconfiguration is described as the modification of an existing smart product in order to fulfil new requirements [86]. In Smart PSS, the engineering change has more strengths compared with traditional project management in two aspects, i.e. in-context change management and digitalized service change [38]. In a smart and connected environment, more reliable engineering change decisions can be made in an automatic way based on the real-time product-sensing data and user-generated data in the context (e.g. the usage environment or user behaviour) of the system.

Smart reuse/remanufacturing is only mentioned in [30]. By using RFID tags and sensor technologies, higher reuse values (e.g. decreased cost) can be offered to industries, especially those time-sensitive industries (e.g. fashion industries) [93]. Meanwhile, Smart PSS adoptions facilitate the remanufacturing process and improve the quality of the remanufactured products. In terms of the product lifetime information and the product composition information embedded with the products via RFID, the firm can plan the remanufacturing process before collecting the products [94]. Similarly, the predictability of the remanufactured products can be guaranteed based on the information as well, improving the quality of remanufactured products and decreasing wastes and costs of some tasks (e.g. sorting, disassembly, disposal).

4.5.5. End-of-life stage

Alcayaga et al. [30] introduced a novel concept ‘smart-circular system’ by integrating Smart PSS and circular economy into an overall consideration. *Smart recycling* during end-of-life is one of the core strategies in the system. The product recycling process can be more efficient and effective owing to the advanced IoT technologies and the product lifecycle data management. By attaching sensors and product composition information to the smart products themselves, the design of recycling process (e.g. the process of waste product collection) could be simplified based on the attached material information of products instead of using an extra database [95]. At the same time, the quality of product recycling can also be improved. The service of smart sorting and disassembly, as a Smart PSS proposition with sensors and product composition information, enables higher purity of recycled materials [96]. With new capabilities generated in Smart PSS, the performance of the recycling tasks can be improved, making the product lifecycle become a close loop.

4.6. Business innovation aspect

From business innovation aspect, this section outlines the key business models in Smart PSS, types of digital platforms, value co-creation process and the capabilities towards a circular economy.

4.6.1. Business model

Smart PSS represents an advanced digital business model, i.e. digital servitization [97,98]. Nevertheless, it follows the typical classifications defined in [20], including *product-oriented services*, *use-oriented services* and *result-oriented services*, with respective examples shown in Table 6. One can find that only a few cases perform a result-oriented business

Table 6
Eight types of PSS business models.

Business models	Types	Example	PSS description	References
Product-oriented	Product-related services: “the provider not only sells a product, but also offers services that are needed during the use phase of the product.”	Smart wearable mask	The company sells the mask and provides the service of performance optimization according to the usage status	[15]
		Smart appliances	The service provider sells the smart appliances together with maintenance and other information services (e.g. product lifetime estimation)	[71]
		Teorema for Carpigiani	Smart appliance solution is implemented which can offer machine remote control, machine management and machine web interface	[71]
		Smart ordering system	The smart ordering system allows tracking and previewing of the personalized configuration before postponed manufacturing process	[37]
	Advice and consultancy: “in relation to the product sold, the provider gives advice on its most efficient use.”	Worry-Free Mobility for Electric Vehicles	Battery mobility advisor and battery life management analytics to improve the product for re-design purposes thereby bridging the battery back to business (Battery to Business)	[72]
Use-oriented	Product lease: “The provider has ownership, and is also often responsible for maintenance, repair and control. The lessee pays a regular fee for the use of the product”	Xerox Printer and Copier Rental	Providing world-class portfolio of copiers, printers, multifunction devices and production equipment to customers and a team of specialists to help with planning, installation, configuration and operation for the printer and copier rental	[100]
	Renting or sharing: “The same product is sequentially used by different users.”	Rolls Royce “Power-by-the-Hour” service Car2Go	A new service offering harnesses the power of Big Data to monitor, plan and perform maintenance and repairs on all the equipment installed on the cargo vessels Offering car rental services but users can book the car on apps and leave the car anywhere on the street in the operating area	[101] [102]
	Product pooling: “a simultaneous use of the product”	3D Hubs	A cloud-based on-demand 3D printing service with a shared pool of manufacturing resources	[103]
	Activity management/outsourcing: “a part of an activity of a company is outsourced to a third party.”	Bosch production line performance Smart box	Bosch crowdsources the Big Data analytics task of smart production line performance to the general public to seek best solution Cloud-based logistics with smart boxes for on-demand delivery	[104] [78]
Result-oriented	Pay per service unit: “The PSS still has a fairly common product as a basis, but the user no longer buys the product, only the output of the product according to the level of use.” Functional result: “for a functional result in rather abstract terms, which is not directly related to a specific technological system. The provider is, in principle, completely free as to how to deliver the result.”	/		

Table 7
Classification of digital platforms.

Categories	Definition	Types	Author, Year	Examples
Product-dependent	An IoT-enabled modular platform consisting of various digitized physical products and its dependent smart e-services.	Product re-/configuration, remanufacturing, and recycling system	Abramovici, Göbel, and Savarino (2017) [86] Savarino et al. (2018) [88] Zheng et al. (2018a) [28] Hasselblatt et al. (2018) [116] Alcayaga, Wiener, and Hansen (2019) [30] Zheng, Chen, and Shang (2019) [38] Benade et al. (2016) [109] Zheng, Xu, and Chen (2018) [117] Lee, Kao, and Yang (2014) [49] Lee, Bagheri, and Kao (2015) [118] Schroeder et al. (2016) [119] Schleich et al. (2017) [67] Feliciano et al. (2017) [68] Tao and Qi (2017) [120] Uhlemann, Lehmann, and Steinhilper (2017) [75] Söderberg et al. (2017) [77] Tao et al. (2018) [82] Zheng, Xu, and Chen (2018) [117] Thomas, Auto, and Gann (2014) [23] Lee and Kao (2014) [121] Baines and Lightfoot (2014) [122] Weiß, Kölmel, and Bulander (2016) [123] Zhang et al. (2016) [78] Zheng et al. (2017) [13] Cenamor, Sjödin, Parida [105] Zheng et al. (2018b) [15] Liu et al. (2018, 2019) [39,108] West, Gaiardelli, Rapaccini (2018) [110] Lee, Chen, and Trappey (2019) [55]	Configurable components of bicycle; reconfiguration of automobile components; remanufacturing of engine components, and etc.
		Embedded open toolkits		Self-adaptable shoes or wearables
		CPS/Digital Twin		Monitoring, control and optimization of machine tools; prognostic health management; high-fidelity design simulation, and etc.
Product-independent	An IoT-enabled modular platform consisting of various smart e-services, which are independent with the composition of physical products.	Service platform		Smart logistics arrangement; smart cooking advice; manufacturing capability planning; smart warning of breathing condition, and etc.

Table 8
Key aspects of value co-creation in literature.

ID	References	Actors	Co-creation stages	Mechanisms	Open innovation level
1	Valencia et al. (2014, 2015) [10,11]	Service provider and consumers	Co-design and evaluate	Perceptive, Responsive	System level
2	Takenaka, et al. (2016) [71]	Service provider and consumers	Co-evaluate	Responsive	System level
3	Weiß, Kölmel, and Bulander (2016) [123]	Suppliers and manufacturing companies	Co-implement	Perceptive	System-of-Systems level
4	Benade et al. (2016) [109]	User and designer	Co-design and implement	Perceptive	Product-Service level
5	Marlungo et al. (2016) [124]	Platform provider (manufacturer) and end users	Co-implement	Perceptive, Responsive	System-of-Systems level
6	Grubic and Peppard (2016) [125]	Manufacturer, service provider and customer	Co-evaluation	Perceptive	System level
7	Lenka, Parida, and Vincent (2017) [43]	Manufacturer, service provider and customer	–	Perceptive, Responsive	–
8	Li and Found (2017) [126]	Suppliers, customers, end users and the society	–	–	System-of-Systems level
9	Liu et al. (2018, 2019) [39,108]	Manufacturer, service provider and customer	Co-design, implement and evaluate	Perceptive, Responsive	System-of-Systems level
10	West, Galarelli, and Rapaccini (2018) [110]	Platform provider (manufacturer) and end users/suppliers	Co-implement	Perceptive, Responsive	System-of-Systems level
11	Chowdhury, Haftor, and Pashkevich (2018) [14]	Manufacturer, service provider and customer	Co-design, implement and evaluate	–	–
12	Zheng P, et al. (2018a, 2018b) [15,28]	Manufacturer/designer, service provider, and users	Co-design, implement and evaluate	Perceptive, Responsive	Product-Service level & System level

Table 9
Digitalization capabilities for the circular economy.

Lifecycle stage	Functionality	Digitalization capabilities			Circular economy value driver		
		Connect capability	Intelligence capability	Analytic capability	Increased resource efficiency	Extended Lifespan	Closed the Loop
Design stage	Adaptable SCP design	✓	✓	✓		✓	✓
Manufacturing stage	Service innovation	✓	✓	✓		✓	
	Resource management	✓	✓	✓	✓		
	Production planning	✓	✓	✓	✓		
Distribution/logistic stage	Production process control/reconfiguration	✓	✓	✓	✓		
	Logistics and packaging	✓	✓	✓	✓		
	Geometry assurance	✓	✓	✓	✓		
Usage stage	Remote monitoring	✓	✓			✓	
	Predictive maintenance	✓	✓	✓		✓	
	Engineering change management	✓	✓	✓	✓	✓	
End-of-life stage	Product reconfiguration	✓	✓	✓		✓	
	Product reuse	✓	✓	✓		✓	✓
	Product remanufacturing	✓	✓	✓		✓	✓
	Product recycling	✓	✓	✓			

Table 10
Examples of application scenarios in Smart PSS.

Sectors	Application scenarios	Reference	Description
Smart manufacturing	Digital-twin-enabled shaft manufacturing	[48]	(1) devise plan of allocating steel bars and CNC machines; (2) generate and simulate production plan; and (3) evaluate production plan throughout the manufacturing process
	GPS-based machine tool prognostics and health management (PHM) Watchdog agent	[29]	Holistic data analysis and health condition monitor services of machine tools based on cyber-physical systems with 5C architecture
	A data-driven personalized wearable mask	[17] [15]	Monitoring machine's performance and make predictive alarming (1) cloud-based breathing condition monitoring (online service innovation); and (2) checking mask fitness (offline service innovation) during usage
Smart living/Smart appliance	Smart water dispenser	[38]	Automatic engineering modules change management
	Cyber-physical smart, connected bicycle	[28]	A CPS-based smart open architecture bike prototype, of which performance analysis, parametric design and predictive maintenance can be achieved
Smart agriculture	Digital-twin-enabled power transformer	[48]	(1) Real-time state monitoring of power transformer; (2) energy consumption of power transformer itself; and (3) output power quality prediction and analysis
	Smart car maintenance and insurance	[37]	(1) Notifying local repair shops nearby based on GPS once the car accident happens; and (2) providing real-time quoting based on text mining and image processing
	Smart remote machinery maintenance systems with Komatsu Digital services for a tractor company	[49] [54]	(1) Remote prognostics and monitoring and (2) remaining life prediction Providing tailor-made services and solutions, for example, the personalized everyday working plan based on data analyzed on previous work shifts
Smart city	Leveraging large-scale smart phones for crowd sensing services	[130]	Urban planning, disaster monitoring/warning etc.
Smart business	Innovation strategies for a mattress company	[72]	Analyzing the current scenarios and innovative scenarios of mattress that whether they satisfy visible and invisible customer requirements and offer out-of-the-box services
	Innovative strategies for an automotive company who produce electric vehicles	[72]	Adopting the proposed innovative matrix for the automobile manufactures for innovation and to achieve 'Worry-free Mobility'
	Disruptive innovation strategies for a small software-oriented company	[123]	Disruptive innovation of a small software-oriented company, by reorientation and creating new service-based offerings for intelligent energy management

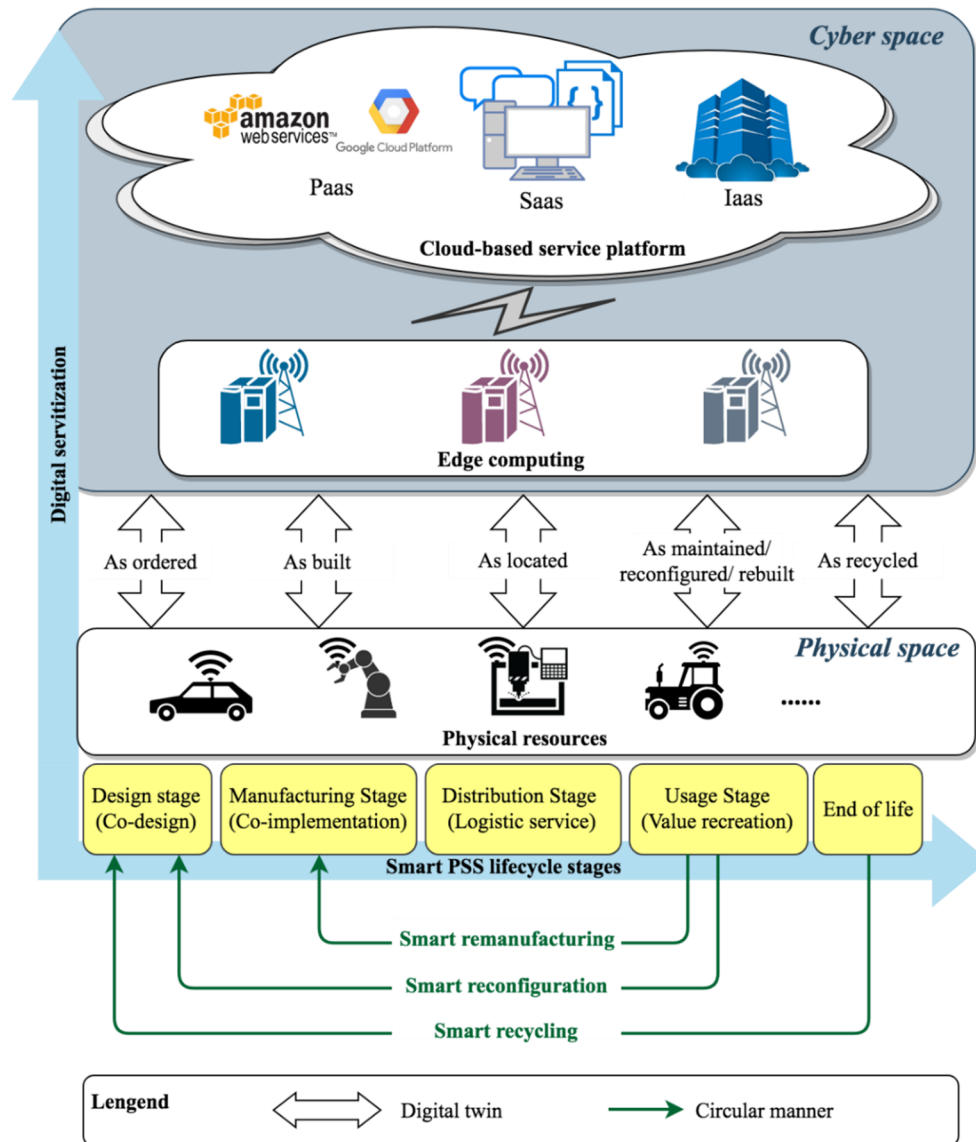


Fig. 9. An overview of closed-loop Smart PSS lifecycle management in a circular economy.

model (software-oriented), while most companies still adopt a product-oriented (e.g. remote diagnosis [81]) or use-oriented one in a shared manner (e.g. cloud manufacturing [99]). This is also in line with the fact that Smart PSS is enabled by the SCPs and its generated services in a servitization process rather than a productization manner. Nevertheless, it is claimed that once the coverage/popularity of affordable SCPs reaches a certain level, they will be only considered as the common tools and media for service generation, and result-oriented business model (e.g. user experience rather than physical products) will be dominant by that time.

4.6.2. Digital platform

As mentioned previously, Smart PSS undertakes a platform-based approach [105], of which can be viewed as an organizational business perspective for leveraging the value of digital technologies based on modularity and IT-enabled interactions [23]. Based on the types of services offered, the authors further classified the digital platforms in Smart PSS into two categories, i.e. *product-dependent digital platform* (digitalized services), and *product-independent digital platforms* (e-services), which are defined and summarized in Table 7. The *product-dependent digital platform* offers modularized digitalized services highly related to the physical product itself, such as product re-/configuration,

real-time monitoring and control. Meanwhile, the *product-independent digital platform* offers those modularized add-on e-services in the service platform which are relative independent with the physical products, such as cooking advices, weather forecasting, etc.

4.6.3. Value co-creation

Smart PSS follows the S-D logic [1], which represents a dynamic, continuing narrative of value co-creation through resource integration and service exchange [106]. This value co-creation process [107,108] can be further depicted by its *interactive actors* (co-exist stakeholders), *stages* (co-design, co-implement, co-evaluate), *mechanism* (perceptive, responsive), and *open innovation levels* (product-service level, system level, system-of-systems level), as shown in Table 8. One can find that in line with the classifications illustrated in Section 4.1, the level of open innovation corresponds to the interactive actors, where user-service provider/designer/manufacture are interacted in the first two levels to derive individualized solutions, while platform providers and manufacturing companies/suppliers interacted in the last level to optimise the overall system impact. Meanwhile, among the co-creation stages, co-design stands for the active engagement of users in the value design process (e.g. smart configuration [109]), co-implement represents the value implementation process through direct interaction and agent

interaction [39] (e.g. platform provider and suppliers [110]), and co-evaluate denotes the evaluation indexes including system-related (e.g. service performance), context-related (e.g. environmental information) and human-related (e.g. user experience) ones. Lastly, enabled by the digital capabilities, the value co-creation mechanism can be classified into *perceptive mechanism* which allows the companies to identify, assess, and address specific customer needs (e.g. customize solutions in virtual space [15]), and *responsive mechanism*, which implies companies react to their customers' emerging and changing needs so that they can participate in the value co-creation quickly and proactively (e.g. dynamic pricing based on usage data logs [71]).

4.6.4. Digitalization capabilities for the circular economy

As mentioned earlier, the ultimate goal of Smart PSS is to meet individual customer needs with sustainability concerns. With a prevailing discussion and tendency towards circular economy [111], it is also interesting to discover the interrelationship between digitalization capabilities and circular economy. The authors extract the key finding in the digitalization capabilities for the circular economy from [30,32,112–115] together with information from Table 5, and summarized in Table 9 with respect to *lifecycle stages*, *functionality*, *digitalization capabilities*, and *circular economy value driver*. One can find that from circular economy value driver perspective, it can be noted that most functionalities help extend product lifespan during design and usage stage (e.g. software upgrade) or achieve better resource efficiency during manufacturing stage (e.g. remote control), while only a few can close the loop during design or end-of-life stages (e.g. smart recycling).

4.7. Application scenarios

Table 10 summaries the typical application sectors, scenarios, and their descriptions of Smart PSS, and it is interesting to find that though most of the applications are still within the smart manufacturing sector [29,48,57,127], however, gradually they are moving to other sectors such as smart living, smart city and smart business as well. Some points in these emerging fields can be summarized as: (1) Smart PSS provides new capabilities for online and offline service innovations of the smart appliance/smart living. The quality of the appliances' performance can be ever improved with higher efficiency and guaranteed with higher flexibility and robustness. (2) Smart PSS also have emerged in the field of smart agriculture, where by integrating the agricultural equipment and relevant information together, the field of agriculture can be a smart ecosystem. (3) Though the concept of smart city (e.g. sensing-as-a-service) has already discussed from the aspect of weather, traffic, lighting, flood and so on, the integration of smart city and Smart PSS provision in a sustainable and customized manner still remains to be investigated. (4) Most of the smart business scenarios discusses about the future business generation. However, there is still a lack of consensus which business innovation method has the capabilities such as generalization and robustness to fit different business features.

5. Challenges

Despite the above achievements in both technical and business aspects, as an emerging paradigm, Smart PSS still faces several challenges in its development and implementation process, which are outlined as three hybrid concerns below.

5.1. Hybrid design concern

Smart PSS takes both design and inverse design process into a close-loop consideration, i.e. *hybrid design*, by utilizing the advanced IT, where informatic-based approaches serve as the key to enable its solution design success. Nevertheless, two challenges remain not well-solved at the moment. Firstly, in the early design phase, other than conventional PSS which provides a pre-defined product-service family

in the web-based platform for user's configuration, Smart PSS should also well-define the smart, connectivity components in the IoT-enabled environment and be adaptable to the dynamic changes of user requirements during usage stage. Secondly, in the usage stage, driven by the massive user-generated data and product-sensed data in the smart, connected environment, Smart PSS should be capable to change/upgrade with context-awareness for redesign innovation/improvement [128].

5.2. Hybrid value concern

Smart PSS is an IT-driven value co-creation business paradigm, and the ultimate goal is to achieve open innovation (social value) with sustainability concerns (environmental value).

5.2.1. Open innovation concerns

In line with the concept of open innovation 2.0 [129], from industrial perspective, Smart PSS can be seen as a human-centric socio-technical ecosystem. Stakeholders become an integral part of the innovation process (i.e. co-creation) by leveraging disruptive technologies, and their experience, i.e. user experience, drives the innovation process. Hence, Smart PSS development and implementation should always put human beings in the loop in an open innovation manner, which is not well-addressed in the existing works.

5.2.2. Sustainability concerns

Smart PSS follows the fundamental principles of PSS, which emphasizes a servitized value proposition with less environment impact for a circular economy [111]. Nevertheless, only a few qualitative studies have discussed its potentials to address the sustainability issues (i.e. increased resource efficiency, extended lifespan, and closed the loop) [114,126], while little in-depth research has been done to truly realize the smart circular PSS [30].

6. Future perspectives

To overcome those challenges, corresponding approaches and future research directions are highlighted in this section to motivate more open discussion and in-depth research in the near future.

6.1. Self-adaptable solution design with context-awareness

To fulfil the hybrid design challenge, product-service design (solution design) should take self-adaptability and context-awareness into an overall consideration. *Self-adaptable design with context awareness* follows the adaptable design principles [131], by taking functional modelling and lifecycle changeability into an overall consideration in the early design stage [132]. Meanwhile, it broadens the scope by adjusting its product or service modules with self-adaptability (i.e. inverse design) in a data-driven manner triggered by the specific context. Moreover, to achieve hybrid design with context awareness, at system level, a solution can be modelled in a product-service-condition family manner $\langle P, C, S \rangle$ that "given a certain context (condition), what products and services should be triggered", where product and service information can be decomposed into different modules/components, and conditions can be easily accessed through hardware sensors (product-sensed data) or social sensors (user-generated data) in an IoT-enabled manner [15].

6.2. Hybrid crowd sensing with cloud-edge computing infrastructure

To ensure hybrid intelligence, effective computing infrastructure, data collection and fusion techniques, and solution knowledge management should be provided, respectively:

Cloud-edge computing infrastructure. Advanced information processing techniques (e.g. deep learning) and increasing hardware computing

capability (e.g. CPS) enable ones to take full advantages of the heterogeneous data/information sensed from various devices and generated from massive users to create many intelligent services. In this context, end devices are not only data consumers from the cloud, but also data producers performing computing offloading, data storage, caching and processing, as well as distribute request and delivery service from cloud to users [133]. By leveraging the advantages of both cloud computing and mobile edge computing, a cloud-edge computing [134] infrastructure should be implemented in the Smart PSS environment.

Hybrid crowd sensing technique. In the industrial environment, both the high reliability and performance of static sensing nodes/infrastructure, and the large-scale, cost-effective mobile devices with high mobility should be leveraged. Moreover, other types of reliable data sources (e.g. product-service information, service records, etc.) should be considered as well, as a hybrid crowd sensing approach, to generate new values.

Graph-based knowledge management. In the Smart PSS environment, 4 V (high volume, variety, velocity and veracity) data sources are collected and converged, consisting of unstructured, semi-structured and structured datasets. Hence, graph database (Nosql database) and graph-based techniques (e.g. graph embedding) should be deployed for the data/information storage, knowledge representation and reasoning process.

6.3. Human-centric Smart PSS

In the Smart PSS context, value co-creation process for open innovation can be modelled as the mapping between design solution space (DSS) and the user solution space (USS) [28,55]. Both spaces are expanded jointly through design iteration, where platform-based approaches serve as the key. Hence, massive users are enabled to contribute their data/information through various channels (e.g. mobile social network), so that valuable knowledge and users' cognitive and affective needs can be extracted readily for value generation purposes.

6.4. An overview of closed-loop Smart PSS lifecycle management in a circular economy

Based on the above future perspectives, Fig. 9 depicts the overview of the ultimate goal of Smart PSS, as a closed-loop lifecycle management system in the circular economy.

The *vertical arrow* (in blue) represents the digital servitization process enabled by the advanced IT infrastructure (i.e. cloud-edge computing) and digital technologies (e.g. CPS, Big Data). Hence, the digital-twin of Smart PSS development (i.e. two-way arrow) can be established in-between the physical and cyber space in a cost-efficient manner based on the analytics of massive user/manufacture/product-generated data and the real-time communication between them (e.g. monitoring, control, optimization, and autonomous). Small-scale, short-term, local data are processed in the edge computing layer, while large-scale, long-term, distributed data can be stored, analysed and retrieved in the cloud computing layer by service providers for decision making, and further pushed to the edge layer and end devices for its service delivery. Meanwhile, the *horizontal arrow* (in blue) presents the different implementation stages of Smart PSS along the engineering lifecycle, of which the digital twin established can enable the design as ordered, manufacturing as built, distribution as located, usage as maintained/reconfigured/re-built, and end-of-life as recycled. Each yellow box stands for a user involved stage, where different stakeholders engage in a value generation manner. The green arrow represents the key aspects to achieve the circular economy, including smart reconfiguration (e.g. automatic engineering change management), smart remanufacturing (e.g. predictive maintenance) and smart recycling (e.g. smart recovery or end-of-life prediction), which further works should be mostly emphasized so as to extend the lifecycle, increase resource efficiency and close-the-loop.

7. Conclusions

Smart PSS, as an emerging IT-driven value co-creation business strategy coined in 2014, has attracted ever increasing attentions among academics recently. Nevertheless, many issues have not been well-addressed and there still lacks a unified/fundamental understanding of Smart PSS so as to distinguish it from other types of PSS. To bridge the gap, this research work provides a systematic literature review of existing works published in the Smart PSS field ever since its first coined in 2014. The major findings can be summarized into three aspects below.

Technical aspect. From Smart PSS development perspectives, it should be conducted in a data-driven platform-based approach with context-awareness. Hence, both human and machine intelligence should be converged in the context. Meanwhile, both solution design and inverse design should be integrated into an overall consideration. From implementation perspectives, Smart PSS has been widely implemented in various application scenarios such as smart manufacturing, smart city and smart living, and different stages of engineering lifecycle, including design stage (co-design), manufacturing stage (smart manufacturing), distribution stage (smart logistics) and usage stage (inverse design/reconfiguration/predictive maintenance). Nevertheless, end-of-life stage (e.g. remanufacturing/recycling) is rarely discussed in literature to enable the closed-loop Smart PSS lifecycle. From IT perspective, IoT serves as the foundation of Smart PSS to enable the ubiquitous connectivity, cloud-edge computing provides the cost-effective information processing, CPS and AR assure its digitalization process (e.g. digital twin), AI techniques (e.g. knowledge-based engineering, machine learning), and Big Data analytics ensure the successful business intelligence decision making.

Business aspect. From value proposition perspective, Smart PSS reflects a unique type of digital servitization, where SCPs act as the media and tool to generate e-services including both product-dependent services and product independent ones. Meanwhile, most existing Smart PSS still undertakes a product-oriented or use-oriented manner, while little concerns the result-oriented business model. From value creation perspective, stakeholders are actively engaged in a value co-creation process through the service-platform (e.g. mobile APP) in the Smart PSS context, where both service providers and users contribute in an open innovation environment. From managerial implementation perspective, companies should not only consider digitalization capabilities (i.e. connectivity, intelligence and analytics), but also the product-service innovation, value chain process, organizational changes, human resources and customer relationship management into an overall consideration.

Social and environmental aspect. From social perspective, Smart PSS can be seen as an ecosystem-centric view of open innovation by utilizing disruptive technologies, to solve societal challenges profitably and sustainably (e.g. smart city, smart living). It is a vision of shared value, sustainable prosperity and improvements in human well-beings. Meanwhile, human beings are actively involved in the value generation loop, and their experience, i.e. user experience, and knowledge drives the innovation. From environmental aspect, the ultimate goal of servitization is to have less environmental impact so as to achieve sustainability in a circular economy. It has been revealed that Smart PSS have great potentials to address the sustainability issues (i.e. increased resource efficiency, extended lifespan, and closed the loop). Nevertheless, more quantitative methods, in-depth analysis and implementation cases should be provided to achieve the smart circular PSS.

The authors hope this research can be regarded as the baseline to invite more open research and discussions in the above-mentioned aspects, and also provide useful insights to both academics and industries in their exploration and implementation of Smart PSS in the near future.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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